

ESE_December 2020_COMP_DiscreteMathematics_2UC C305_III_30DEC2020_2:30- 3:00_Objective Questions (20 Marks)

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Mark only one oval.

☐ Div-A

☐ Div-B

Questions

5. Which of the following statement is a proposition?

1 point

Mark only one oval.

- ☐ Get me a glass of milkshake
- ☐ God bless you!
- ☐ The only odd prime number is 2
- ☐ What is the time now?

6. $*$: $A \times A \rightarrow A$ is said to be binary operation if _____.

1 point

Mark only one oval.

- ☐ $a*b$ belongs to A for all a belongs to A
- ☐ $a*b$ belongs to A for some b belongs to A
- ☐ $a*b$ belongs to A for some a, b belongs to A
- ☐ $a*b$ belongs to A for all a, b belongs to A

7. _____ is not a binary operation on $\mathbb{Z}$.

1 point

Mark only one oval.

- ☐ $+$
- ☐ $-$
- ☐ $*$
- ☐ $/$

8. $(S, *)$ is said to be semigroup if _____.

1 point

Mark only one oval.

- ☐ $*$ is binary operation on S
- ☐ $*$ is associative on s
- ☐ S has an identity element with respect to $*$
- ☐ $*$ is binary operation and associative on S both.

9. Let $(G, *)$ be a group then for all a, b in G $(a*b)^{-1} = ?$

1 point

Mark only one oval.

$$a^{-1} * b^{-1}$$

☐ Option 1

$$a^{-1} * b$$

☐ Option 2

$$a * b$$

☐ Option 3

$$b^{-1} * a^{-1}$$

☐ Option 4

10. Which of the following is true about Handshaking Lemma?

1 point

Mark only one oval.

☐ In Handshaking lemma, If the degree of a vertex is even, the vertex is called an even vertex

☐ The degree of a graph is the largest vertex degree of that graph.

☐ The degree of a vertex is odd, the vertex is called an odd vertex.

☐ The sum of all the degrees of all the vertices is equal to twice the number of edges.

11. A graph with one vertex and no edges is _____.

1 point

Mark only one oval.

- ☐ multigraph
- ☐ digraph
- ☐ isolated graph
- ☐ trivial graph

12. The power set of an empty set is?

1 point

Mark only one oval.

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ empty set

13. The relation $R=\{(a,b),(b,a)\}$ on set $X=\{a,b\}$ is?

1 point

Mark only one oval.

- ☐ Irreflexive
- ☐ Reflexive
- ☐ Anti-Symmetric
- ☐ None of the mentioned

14. Let $G=(\mathbb{Z}_6, +_6)$ is an Abelian group then the inverse element of 4 is _____.

1 point

Mark only one oval.

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ 3

15. The number of edges in a complete graph with 'n' vertices is equal to: 1 point

Mark only one oval.

- ☐ $n(n-1)$
- ☐ $n(n-1)/2$
- ☐ n^2
- ☐ $2n-1$

16. Determine which is correct statement with respect to set A and $P(A)$ 1 point

Mark only one oval.

- ☐ $A \cup P(A) = P(A)$
- ☐ $A \cap P(A) = A$
- ☐ $P(A) - \{A\} = P(A)$
- ☐ None of the mentioned

17. The maximum degree of any vertex in a simple graph with n vertices is 1 point

Mark only one oval.

- ☐ $n-1$
- ☐ $n+1$
- ☐ $2n-1$
- ☐ n

18. The transitive closure of a relation R on set A whose relation matrix

1 point

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$
 is :

(A) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

Mark only one oval.

- ☐ A
- ☐ B
- ☐ C
- ☐ D

19. Which of the following statement is not valid for poset.

1 point

Mark only one oval.

- ☐ A nonempty finite poset has a maximal element.
- ☐ A nonempty finite poset has a greatest element
- ☐ A nonempty finite poset has a minimal element
- ☐ None of the mentioned

20. Every linear graph on n vertices must have _____ edges.

1 point

Mark only one oval.

- ☐ $(2(1)+2(n-2))/2$
- ☐ $2n$
- ☐ $2n-1$
- ☐ n

21. What is the weight of the word (01011) in B^5 .

1 point

Mark only one oval.

- ☐ 3
- ☐ 2
- ☐ 5
- ☐ 1

22. Which of the following statement is negation of the statement "2 is even or -3 is negative"?

1 point

Mark only one oval.

- ☐ 2 is even or -3 is not negative
- ☐ 2 is odd or -3 is not negative
- ☐ 2 is even and -3 is not negative
- ☐ 2 is odd and -3 is not negative

23. Let A: All badminton player are good sportsperson. B: All person who plays cricket are good sportsperson. Let X denotes set of all badminton players, Y of all cricket players, Z of all good sportsperson. Then which of the following statements is correct?

1 point

Mark only one oval.

- ☐ Z contains both X and Y
- ☐ X contains Y and Z
- ☐ Z contains X and Y is outside
- ☐ Z contains Y and X is outside

24. In the principle of mathematical induction, which of the following steps is mandatory? 1 point

Mark only one oval.

- ☐ induction hypothesis
- ☐ inductive reference
- ☐ induction set assumption
- ☐ minimal set representation

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